

Introduction to Information Technology

CSIT — 1st Semester — 2081 — Answer Sheet
Subject Code: CSC114

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain different types of software with their roles. How do system software and application software interact with each other to provide a seamless computing experience?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Describe different network topologies along with their advantages and disadvantages. Explain how the networking transformed sectors such as education, business, and healthcare.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the concept of memory hierarchy. How does the hierarchy affect the performance of a computer system? Explain the difference between hardcopy and softcopy output devices with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. List out the characteristics of a computer. How do these characteristics make computers efficient for various applications?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Compare and contrast primary memory and secondary memory.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Describe the different phases of the instruction cycle. How does it help in executing a program?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Convert the binary number 110101.101 to its equivalent octal and hexadecimal representations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Describe the different layers in the internet architecture and how each layer contributes to internet communication.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is database? Explain how databases support operations in business, social media, finance and healthcare.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain how multimedia applications enhance user experience in areas like online learning, gaming, and virtual reality.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Why computer security is essential? Explain how are confidentiality and integrity services maintained.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a) Big Data b) E-Commerce

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.